Chitchula Sai Bhuvan

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Education

Amrita Vishwa Vidyapeetam 2025
M.Tech in Computer Science and Engineering **7.9/10**

Sagi Rama Krishnam Raju Engineering College 2024
B.Tech in Electrical and Electronics Engineering **7.8/10**

Experience

Data Analytics and Machine Learning Intern — Fluentgrid Aug'25 – Dec'25

- Built an **end-to-end churn prediction pipeline** covering **data ingestion, cleaning, feature engineering, exploratory data analysis, and supervised machine learning models** to identify high-risk customers.
- Evaluated **baseline and ensemble models** using metrics such as **ROC-AUC, precision-recall, and confusion matrix**, addressing **class imbalance** and prioritizing **recall** for business-critical churn cases.
- Applied **model interpretability techniques including feature importance and SHAP analysis** to explain predictions and uncover key drivers of churn, enabling **data-driven customer retention strategies**.

Projects

Agentic Bug Hunter – AI-Driven Code Bug Detection System — Agentic AI, MCP, Explainable AI
Sept'25 – Dec'25

- Built an **Agentic AI-based code analysis system** using an **MCP (Model Context Protocol) server** to automatically analyze source code, detect potential bugs, and generate explainable diagnostic insights.
- Designed the **MCP server as the orchestration layer** enabling AI agents to interact with code analysis tools via **modular tool interfaces** and a pipeline for code parsing, bug reasoning, normalization, and explanation generation.
- Integrated **structured logging, explainable AI techniques, and a persistent bug knowledge repository**, producing **structured outputs (CSV/JSON)** with code identifiers, fault locations, and human-readable explanations for validation and benchmarking.

LLM-Assisted Time Series Anomaly Detection System — Machine Learning, NLP Dec'25 – Feb'26

- Developed a **hybrid anomaly detection system** using **Isolation Forest** for unsupervised detection and **transformer-based models** for processing multivariate sensor time-series data.
- Built a data pipeline with **Pandas** and **NumPy**, applying feature engineering (rolling mean, standard deviation) and integrated **SHAP** for model explainability.
- Deployed the system using **Docker** and created an interactive **Streamlit** dashboard to visualize anomalies and provide **LLM-powered explanations** for real-time insights.

Security Analysis and Mitigation of MFA Fatigue Attacks (Uber 2022 Case Study) Jan'26

- Designed and developed an adaptive authentication system that uses behavioural signals to assess login risk in real time and enforce dynamic access decisions (ALLOW / VERIFY / BLOCK).
- Built detection mechanisms for MFA fatigue attacks and brute-force attempts using time-based event analysis, rate limiting, and account lockout strategies.
- Implemented a secure backend architecture with FastAPI, JWT authentication (with revocation), bcrypt hashing, and a tamper-resistant audit logging mechanism for tracking security events.

Technical Skills

Programming Languages: Python, SQL
Core Concepts: Data Structures and Algorithms (DSA), Object-Oriented Programming (OOP), DBMS, Operating Systems, Problem Solving
Mathematics: Linear Algebra, Probability & Statistics, Calculus, Optimization Techniques
Machine Learning / AI: Machine Learning, Data Analysis, NLP, Large Language Models (LLMs), Agentic AI, Prompt Engineering, Fine-Tuning, Retrieval-Augmented Generation (RAG)
Libraries/Frameworks: NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn
Tools: Git
Operating Systems: Windows, Linux

Trainings and Certifications

- Microsoft Certified: Azure AI Fundamentals (AZ-900)
- IIT Bombay:e-Yantra competition regional finalist
- Achieved Gold Level in Accenture's iAspire Go for Gold program by completing technical assignments for every learning module